

Course Outline

Institute of Business Administration

CSE 309: Theory of Automata

Fall 2024

Dr. Asma Sanam Larik

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Saturday 11:30--2:15 Main Campus

Course Objective:

This course is about the beautiful contributions and insights that the field computer science has afforded us. We shall specifically focus on the interplay between notions of logical proof and how they link with limits on computation; which in turn has implications on what can be computed and how efficiently in the physical world. The perspective we gain by looking at our world through the computational lens is perhaps the main contribution of theoretical computer science to the human endeavor. The course starts from basic math preliminaries, works its way through historical breakthrough results in the field and then branches off into an investigation of fundamental challenges at the frontiers of theoretical computer science. The course provides opportunities for both: (i) practice in developing a rigorous mathematical argument, and (ii) practice in developing a clear intuitive explanation of a complex argument. During Introduction to Programming and Discrete Math students learn a new language, i.e., Discrete Math. They obtain some practice in reading, writing, speaking and listening to this new language. In a human languages class, a logical progression is to read and understand the existing classic literature in the language. Similarly, we will engage ourselves with some of the classics in theoretical computer science. The course content is spread over three distinct modules. The first deals with what can be computed in principle: computability, the second deals with how efficiently we can compute: 2 complexity and finally we branch off into an investigation of fundamental challenges at the frontiers of theoretical computer science.

Text Book:

The main text for the course is

- **Introduction to the Theory of Computation by Michael Sipser, 3rd edition.**

Grading Policy:

- **Assignments: 15% (3)**
- **Quizzes: 15% (3)**
- **Midterm: 30%**
- **Final: 40%**

Course Schedule:

Week/Date	Topics

Week 1	Introduction to Finite Automata Preliminaries Alphabets, strings, languages, types of proofs Deterministic Finite Automata
Week 2	Designing DFA Regular Languages and Regular Operations Non deterministic Finite Automata Equivalence of DFA and NFA Closure under Regular Operations
Week 3	Pumping Lemma for Regular Languages Introduction to Context-free Languages Designing CFG
Week 4	Pushdown Automata Definition and Examples
Week 5	Turing Machines Definition and Examples
Week 6	Variants of Turing Machine Multi-tape TM Non deterministic TM Enumerators Equivalence with other models
Week 7	Decidable Languages Undecidability Diagonalization Method

	Turing un-recognizable language
Week 8	Measuring Complexity Analyzing Algorithms Class of P and NP Problems
Week 9	NP-Completeness NP- Hard Proofs and Examples

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